INTRODUCTION TO COURSE

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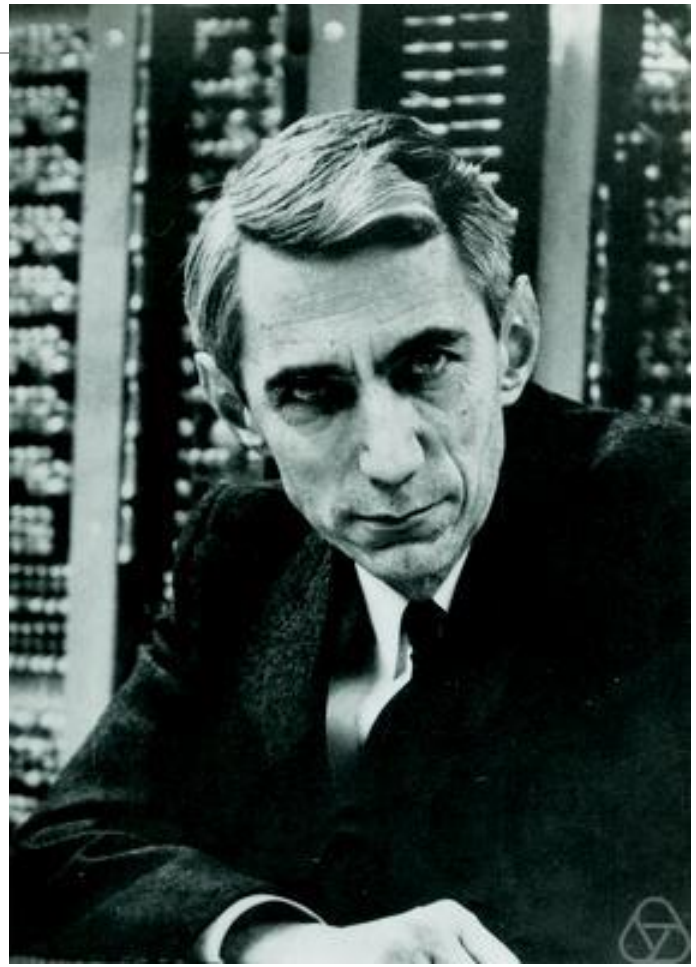
Today's Lecture

- Introduction to digital communication
- Components of a digital communication system
  - Source coding
  - Encryption
  - Channel coding
  - Modulation/demodulation
  - Channel impairments
  - Practical transmitters/receivers

“The fundamental problem of communication is that of reproducing at one point, either exactly or approximately, a message selected at another point.”

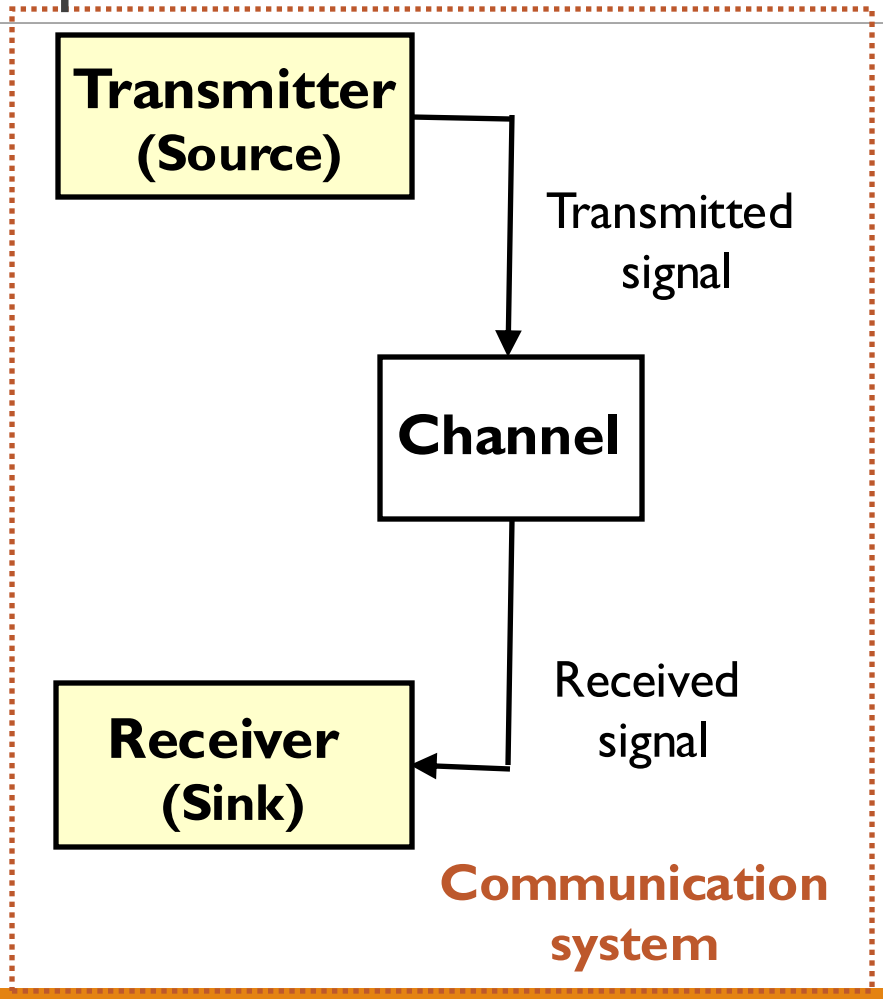
Claude Shannon

A Mathematical Theory of Communication, The Bell System Technical Journal, 1948



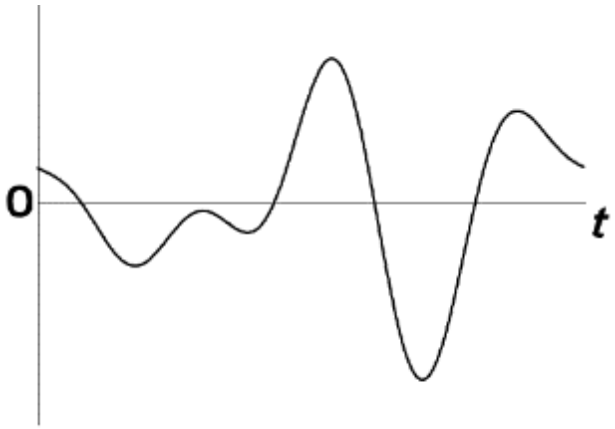
Taken from http://owpdb.mfo.de/

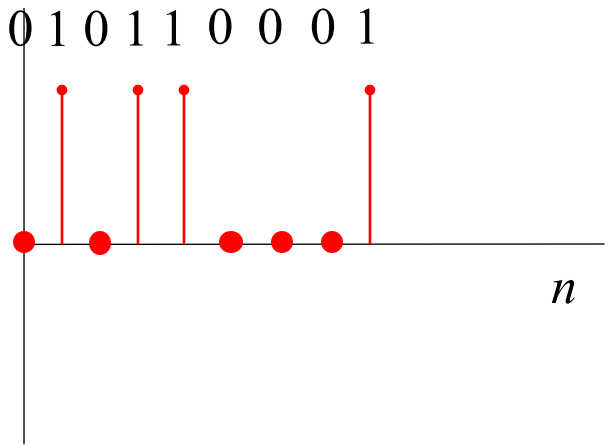
Principles of Communication



- Transmitter: transmits information signal derived from source
- Channel: transfers signal from transmitter to receiver
  - Includes analog circuitry
  - Includes propagation medium
- Receiver: processes received signal to extract the information that was sent

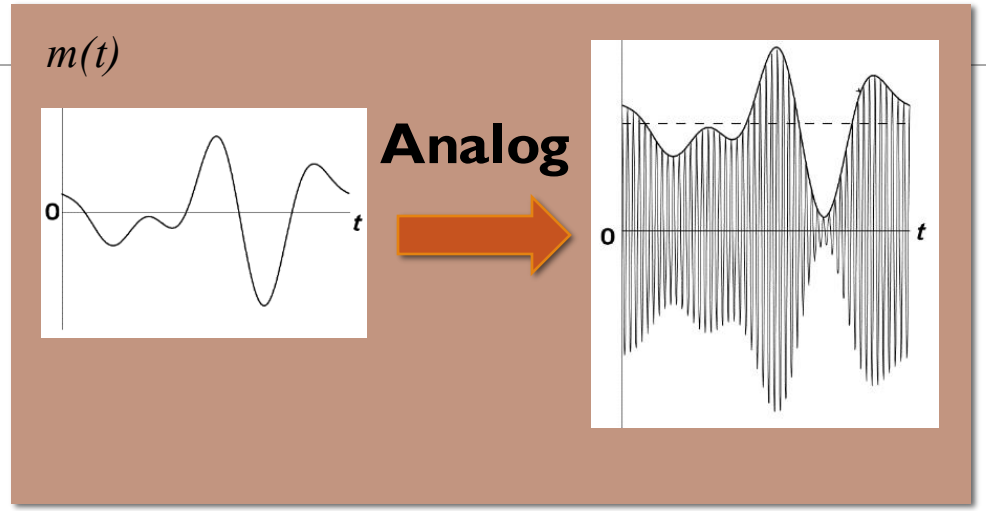
Analog vs Digital Communication

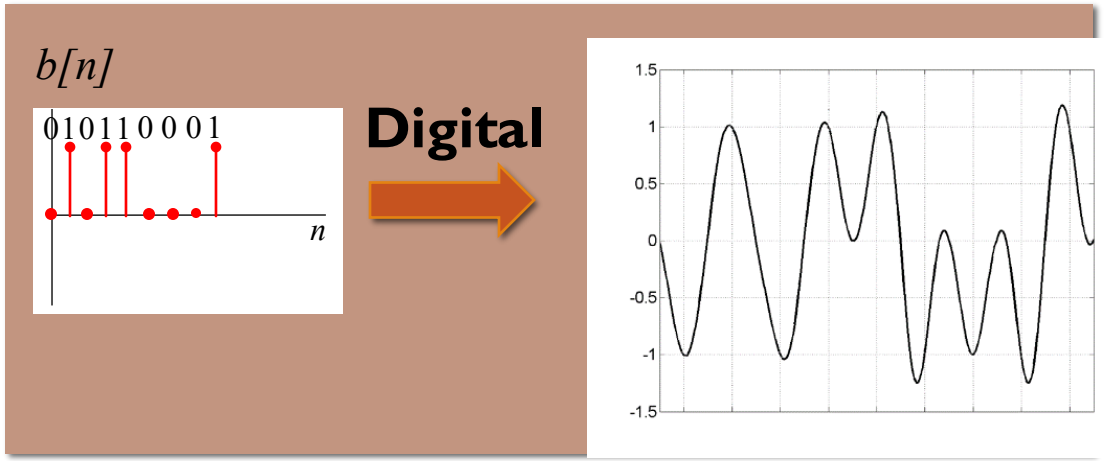




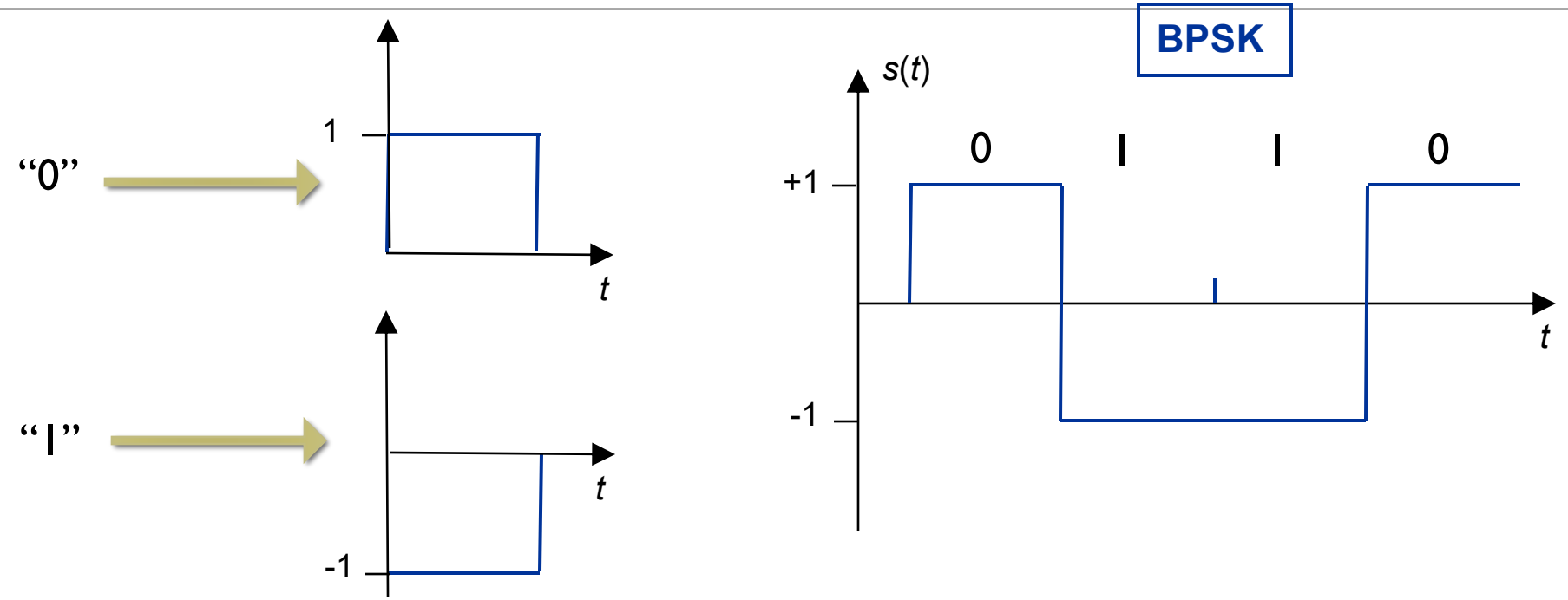
Analog: source is a continuous-time waveform
Digital: source is a digital (binary) sequence
Both systems actually send a **continuous-time wave**

The difference is the source





Analog vs Digital Communication



Digital communication uses a finite number of possible waveforms
Example: BPSK modulation with a rectangular pulse shape

Why Digital Communications

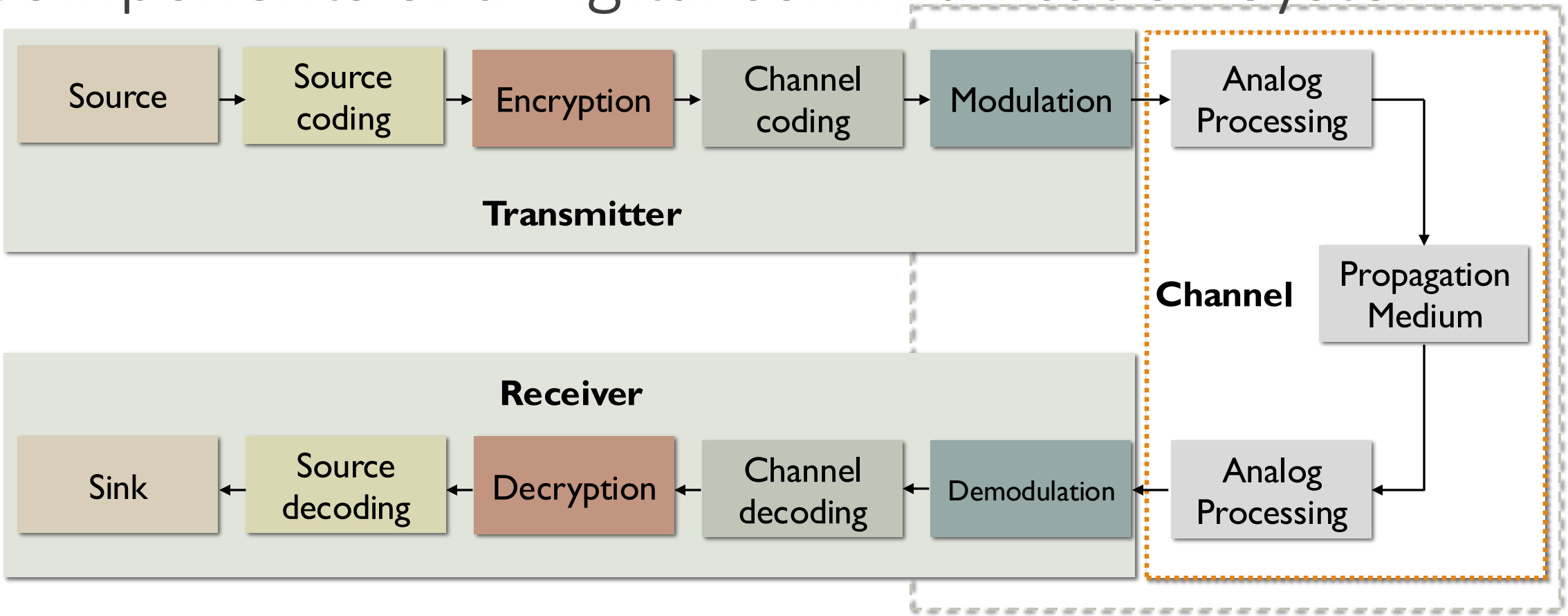
- Suitable for digital data
  - Use high-quality reproducible digital components
  - Analog components have variable specs, effects continuous outputs while digital components are more robust due to discrete levels
- Easier to implement security
- More robust to noise
- Easier to support multiple rates

Why Digital Communications

Easy to share the system with multiple users
Easy to change + reconfigure (e.g., SDR : software defined radio)
Compression of the source data is simpler
- Makes the system more efficient

Uses DSP to take advantage of Moore's law
- The number of transistors per unit area doubles every 18 months
- Reduces power consumption and cost
- Analog components are not advancing as quickly as digital

Components of a Digital communication System



More details will be added throughout the course
Just a reference design, some blocks may be merged or swapped

Source Encoder and Decoder



- **Source**: origin of information
- **Source encoder**: purpose is to perform compression
  - Lossless(e.g. zip encoding)
  - Lossy (e.g. jpeg)
- **Source decoder**: uncompress/reconstruct the original source
  - Perfect reconstruction for lossless codes
  - Perfect reconstruction for lossy codes

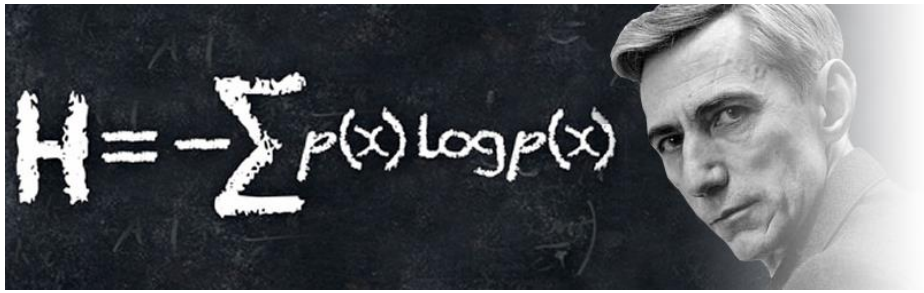
Source encoder and decoder

Entropy for a binary source

Probability for symbol 1

Probability for symbol 2

$$H=-p_1\log_2(p_1)-p_2\log_2(p_2)$$

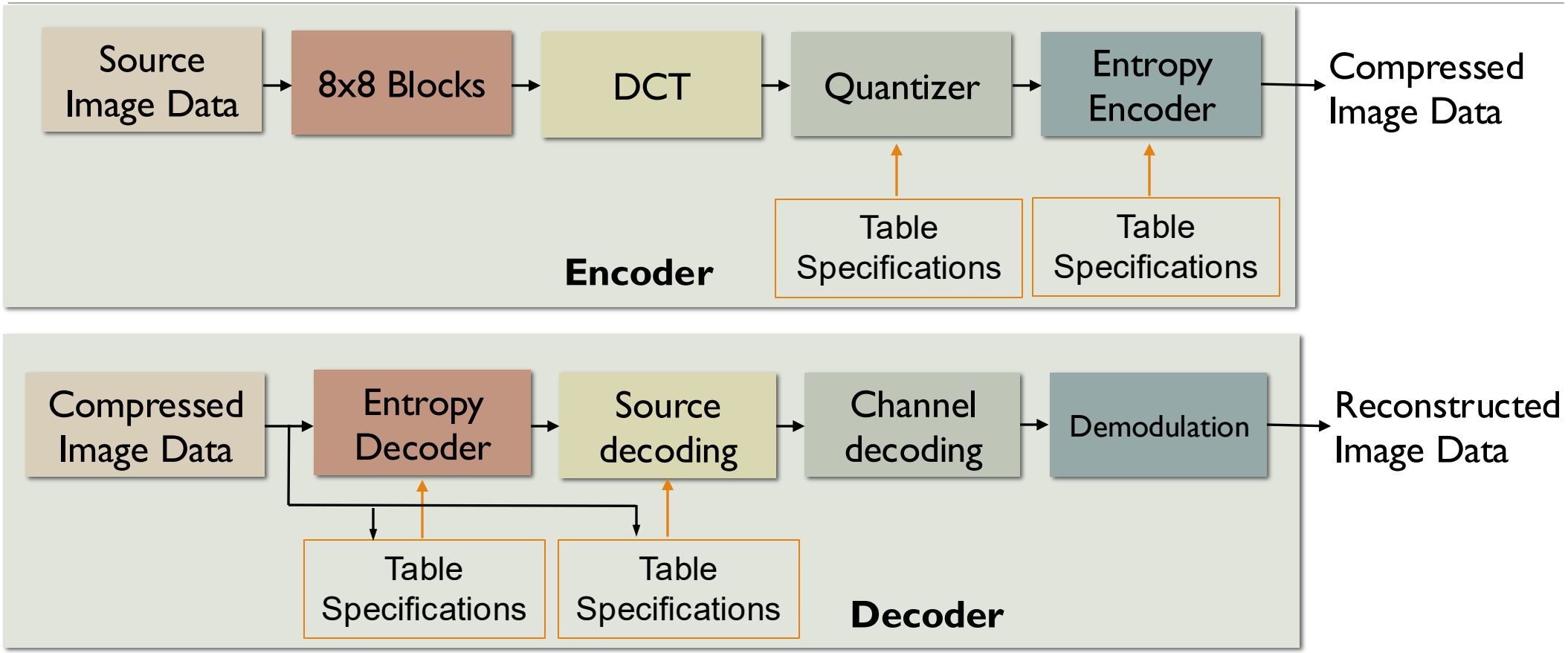


Entropy for a discrete source

- Measures the amount of information contained in a given message
- Average length of the codewords in the best possible lossless data compression algorithm
- Shannon entropy (not Clausius entropy used in thermodynamics)

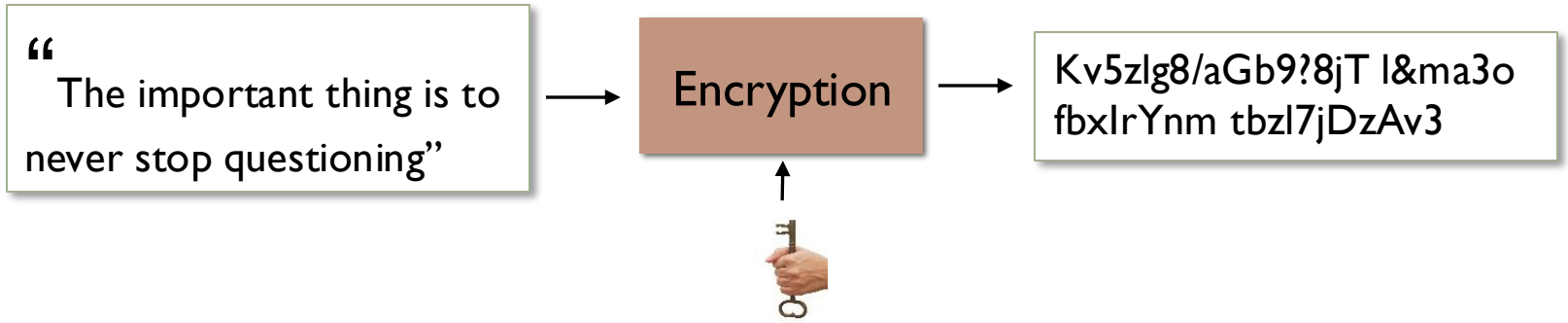
Measures the efficiency of a source coding algorithm

Source encoder and decoder



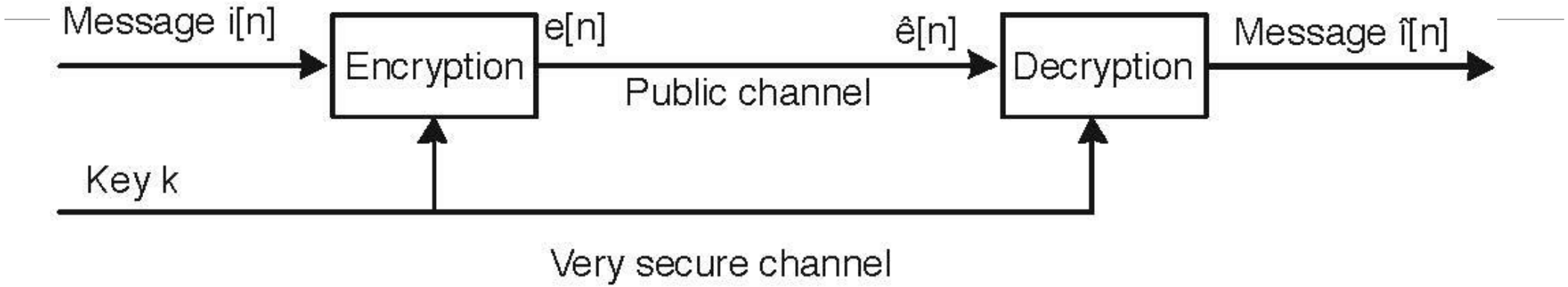
Example: JPEG encoding and decoding processes

Encryption/Decryption



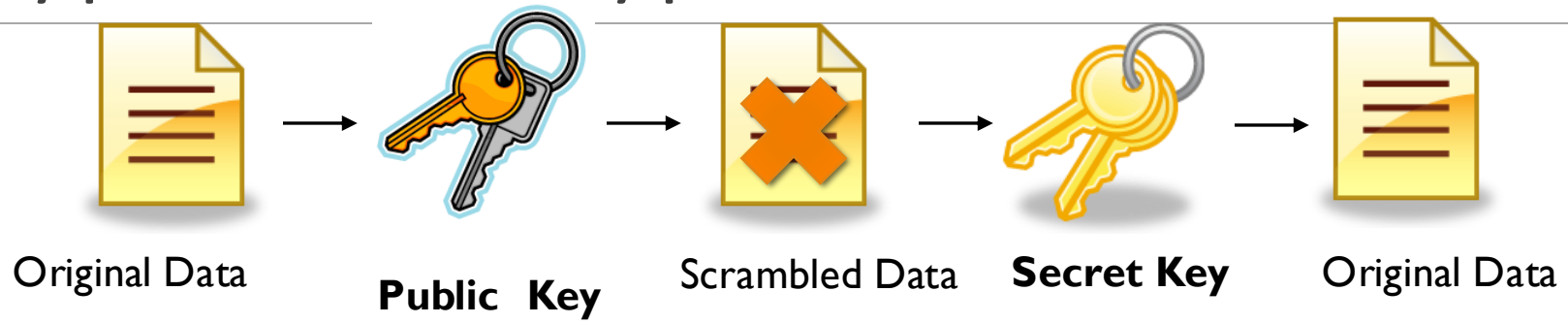
- **Encryption:** convert information into something that is hard to understand by an unintended recipient
  - Component of a security protocol
  - Typical requires exchange of a key
- **Decryption**: remove encryption, make the decoded signal readable

Encryption/Decryption



- Secret Key Encryption
  - A single secret key is used by encryption and decryption
  - *Block ciphers* use the key to encrypt data block by block
    - Used in 3G and 4G
  - *Stream ciphers* generate a stream of pseudorandom keys which are bitwise XORed with the data
    - Used in IEEE 802.11 and Bluetooth

Encryption/Decryption

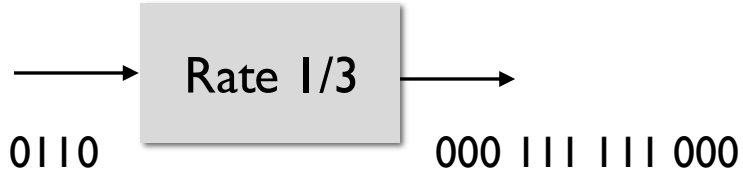


Public Key encryption

- Involves a pair of keys: public key and private key
- Public keys are freely distributed while private keys are secret
- Data encrypted with a public key can be decrypted only with the corresponding private key
- Data encrypted with a private key can be only be decrypted with a public key (called signing, ensure sender is who they claim to be)

Channel Coding

Basic idea: add known redundancy to correct channel errors
Error correction: repair received signal
Example: Repetition code

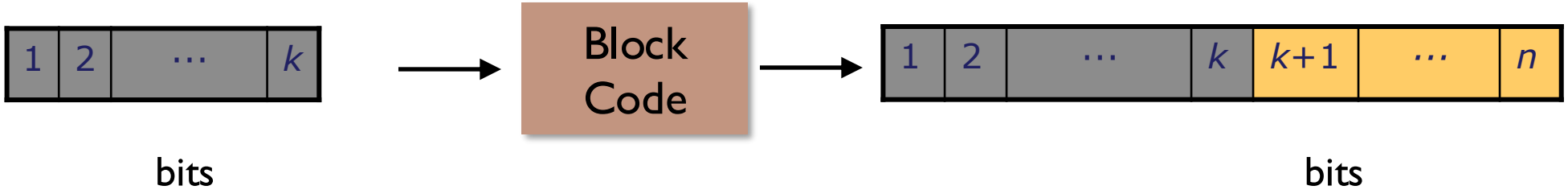


Error detection: determine if there was an error
Example: CRC (Cyclic Redundancy Check)

$$\text{Code rate} = \frac{\#\ of\ uncoded\ bits}{\#\ of\ coded\ bits}$$

Forward Error Correction (FEC) Codes

Also known as Error Controls Codes (ECCs)
Block Codes: developed in 50's

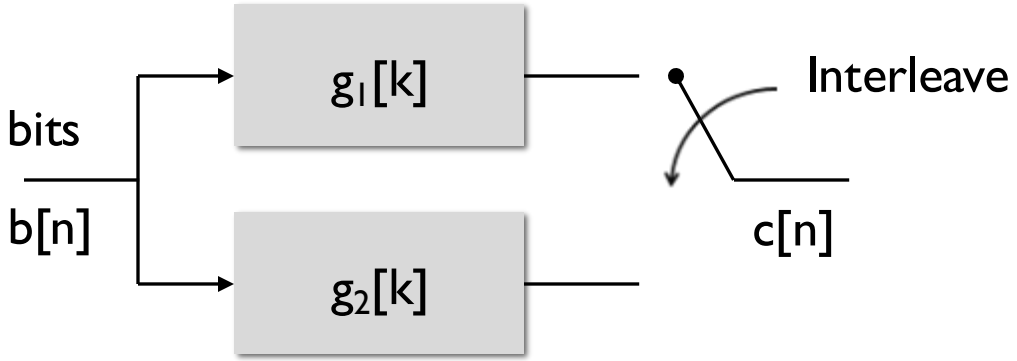


- Example: Rate ½ Block Code, Length 5
  - Block code takes 5 bits in and produces 10 bits out
- Some block codes:
  - CRC code, used in nearly every digital communication protocol
  - Reed-Solomon code, used in xDSL, DVB-S & DVB-S2, IEEE 802.11ad

Forward Error Correction (FEC) Codes

Convolutional Codes: developed in 60's-70's

- Convolve data with multiple impulse responses in binary field



$g_1[k]$ and $g_2[k]$ FIR filters, length $v+1$;
v: constraint length

- Used in GSM, WCDMA, IEEE 802.11a/b/g/n/ac/ad

Forward Error Correction (FEC)

Example: (5,7) rate 1/2 convolutional code

$$c[2n] = i[n] \oplus \mathrm{i}[n\text{-}1]$$

$$c[2n+1] = i[n] \oplus \mathrm{i}[n\text{-}2]$$

- Constraint length of 2 (memory 2 bits)
- More than 2 bits errors is a problem

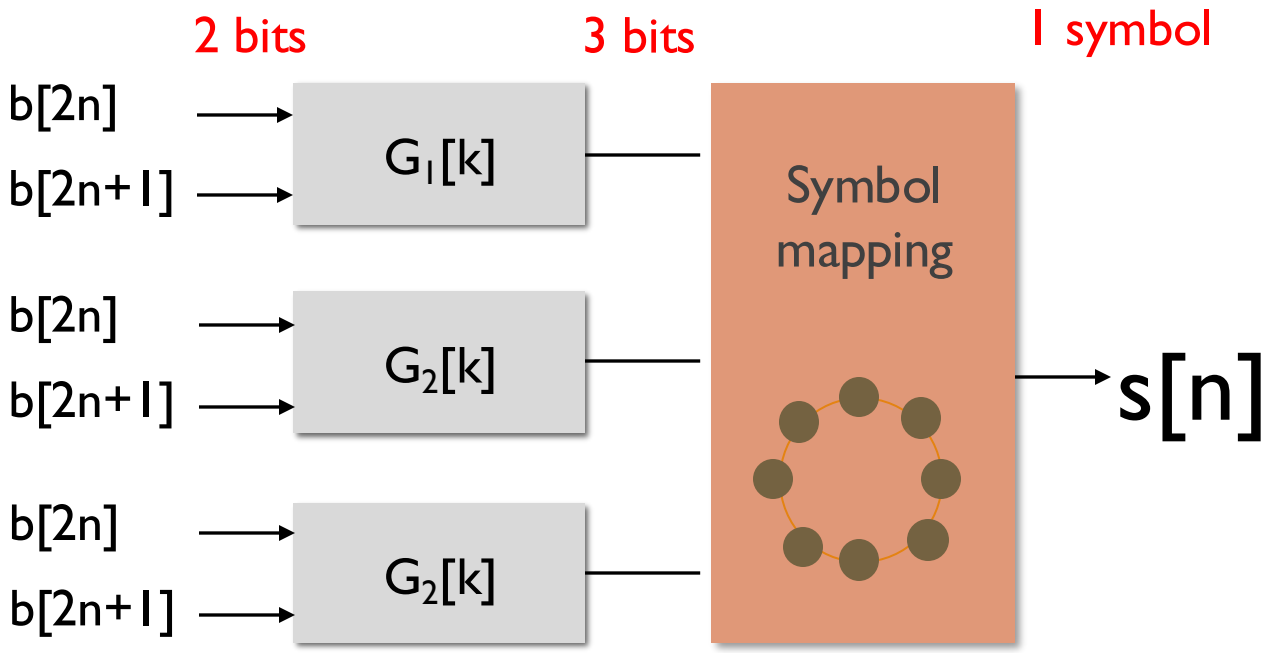
Decoding is important and complex

- Viterbi decoder: finds the data sequence which is closet to the observation according to some distance metric

Forward Error Correction (FEC)

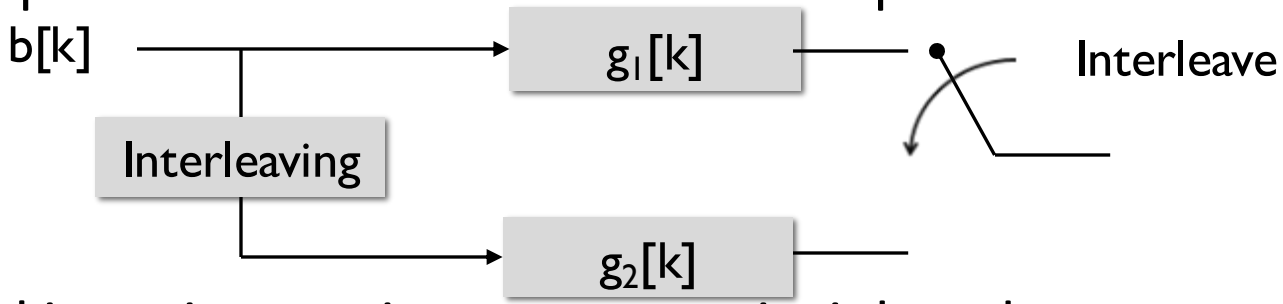
Trellis Codes: developed in 80's

- Generalization of convolutional codes w/ explicit symbol mapping
- Combine modulation and coding
- Used in ATSC HDTV
- Viterbi decoder



Forward Error Correction (FEC)

- **Turbo Code:** invented in ’93, developed in the 90’s - 00’s
  - Tricky extension of convolution coding
  - Uses convolutional code with specially designed IIR filters
  - Uses a deep random interleaver between inputs



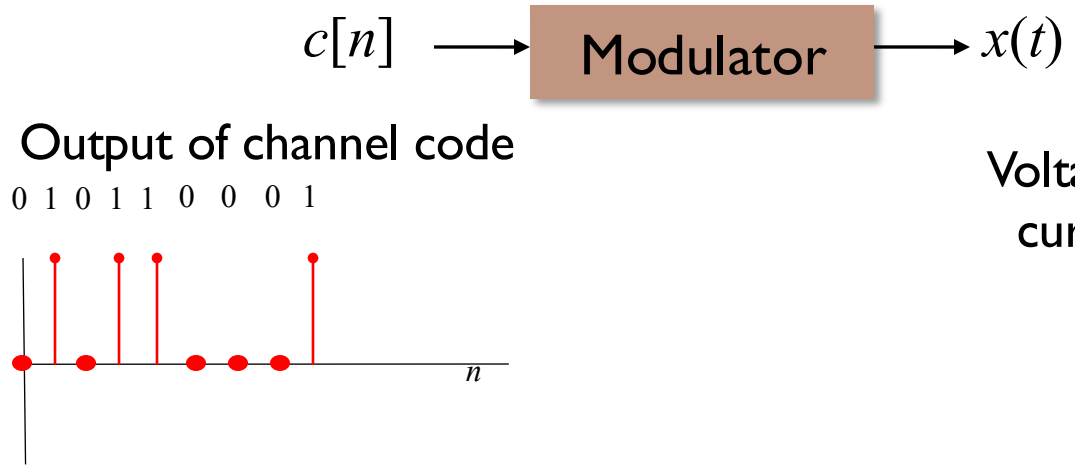
  - Use special iterative maximum a posteriori decoder
  - Very good error protection, but need large block length (e.g., 10,000)
  - Used in the 3G WCDMA and 4G LTE systems

Forward Error Correction (FEC)

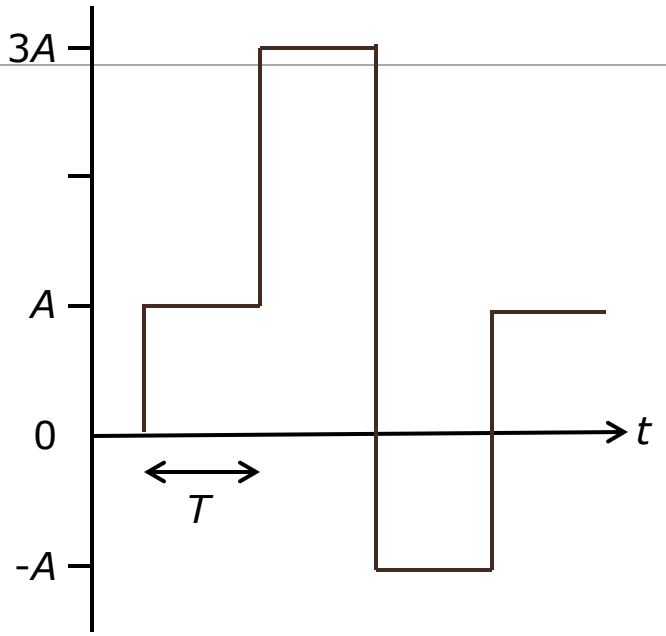
LDPC (low density parity check) code: developed in the 60's, rediscovered in '00s

- Block code with large block length and smart decoder
- Carefully designed combined with special iterative MAP decoder
- Works with smaller block lengths (e.g., ~ 1,000 bits)
- Decoding is more complex
- Used in IEEE 802.11ad, optional in some other standards

Modulation



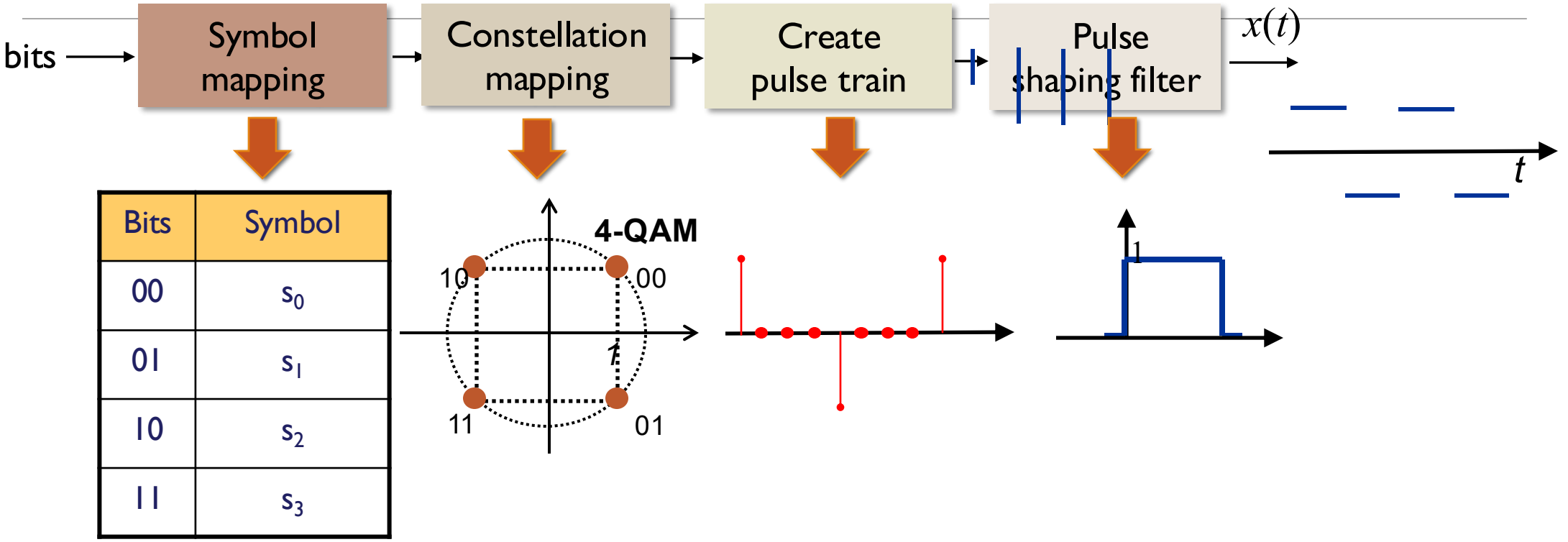
Voltage or current



Maps bits to analog waveforms

- Linear or not linear
- Memoryless or with memory

Baseband modulation



Symbols are mapped to points in constellatic $\mathcal{C} = \{c_1, c_2, \ldots, c_N\}$

- Cardinality: $|C| = 2^k = N$, $k \rightarrow$ # of bits/symbol

Passband Modulator

Time varying amplitude

Carrier frequency

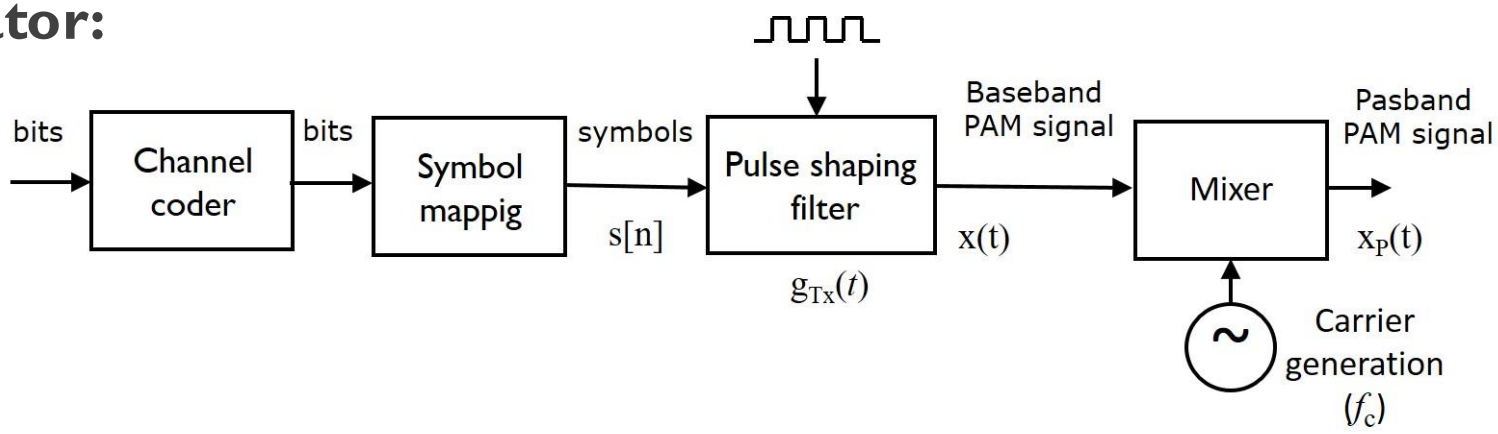
Time varying phase

$$x_{\mathrm{p}}(t) = A(t)\exp(2\pi f_{\mathrm{c}} t + \phi(t))$$

Amplitude, phase or frequency of **RF carrier** varies w/ information

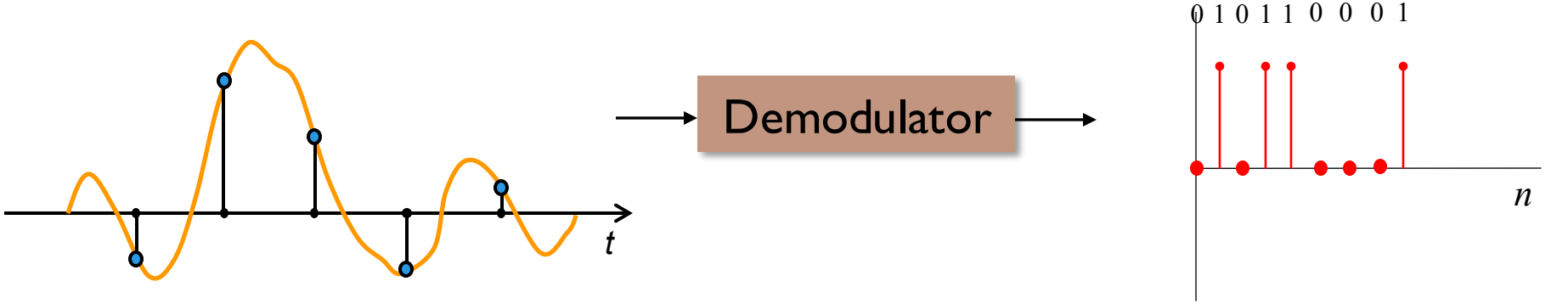
Most common digital passband modulation types: ASK, PSK, FSK M-QAM

Example: ASK modulator:



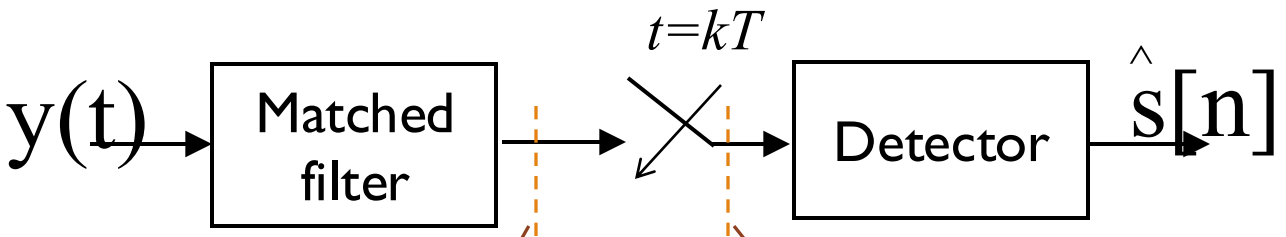
Demodulator

- Use received samples to *detect* transmitted sequence

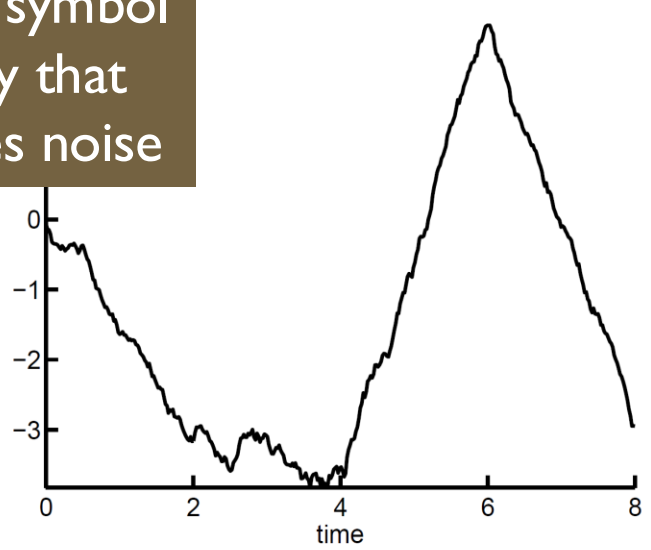


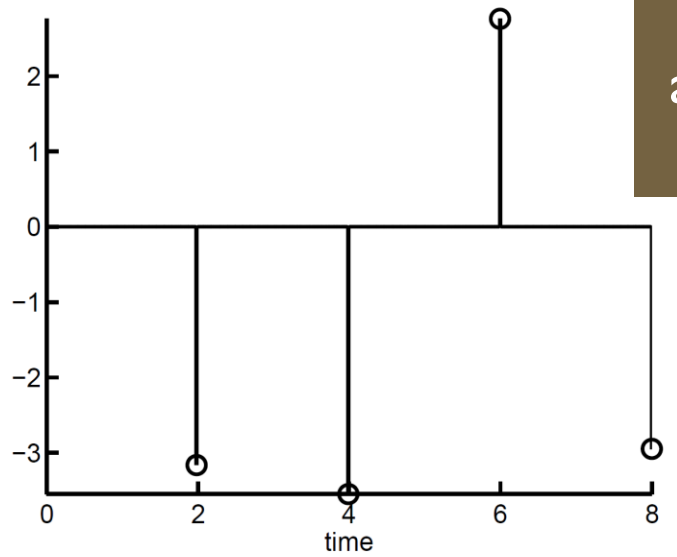
  - May use an estimate of the channel
  - May include equalization
  - May include detection
  - May operate with memory

Basic Demodulation



collects energy of entire symbol in a way that minimizes noise





makes an educated guess about what was sent

Detection Methods

Hard decision: each coded bit is demodulated as 0 or 1

- Only received symbol is used to decide
- *Example:*

Bits	Symbol	Received samples	Slicer output (in bits)
0	1	-0.2	1
1	-1	0.2	0
0	1	0.5	0

Detection Methods

Soft decision: made corresponding to “distance” between received sequence and sequence corresponding to 0 or 1 bit transmission

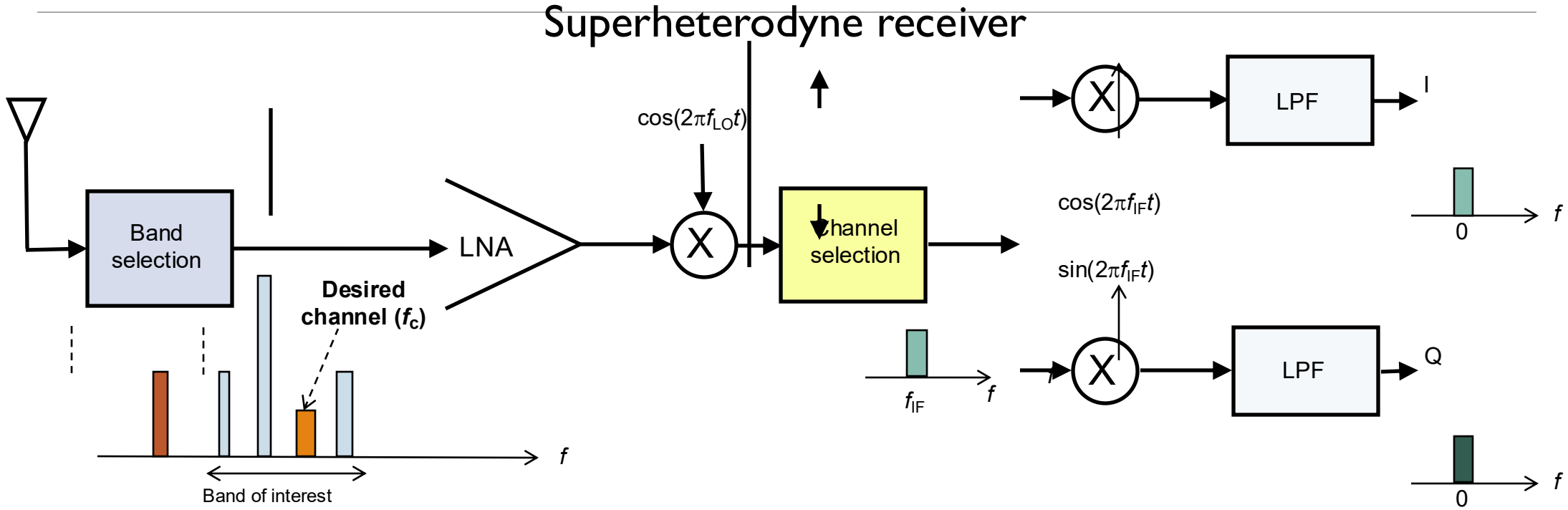
- *Example:*

$$\mathcal{C} = \{(+1, +1, +1), (-1, -1, -1)\}$$

Bits	Symbol	Received samples (RX)	d(RX,(1,1,1))	d(RX,(-1,-1,-1))	Decision
0	(1,1,1)	(1,0.5,-0.5)	$0^2+(1/2)^2+(3/2)^2=2.5$	$2^2+(3/2)^2+(1/2)^2=6.5$	0

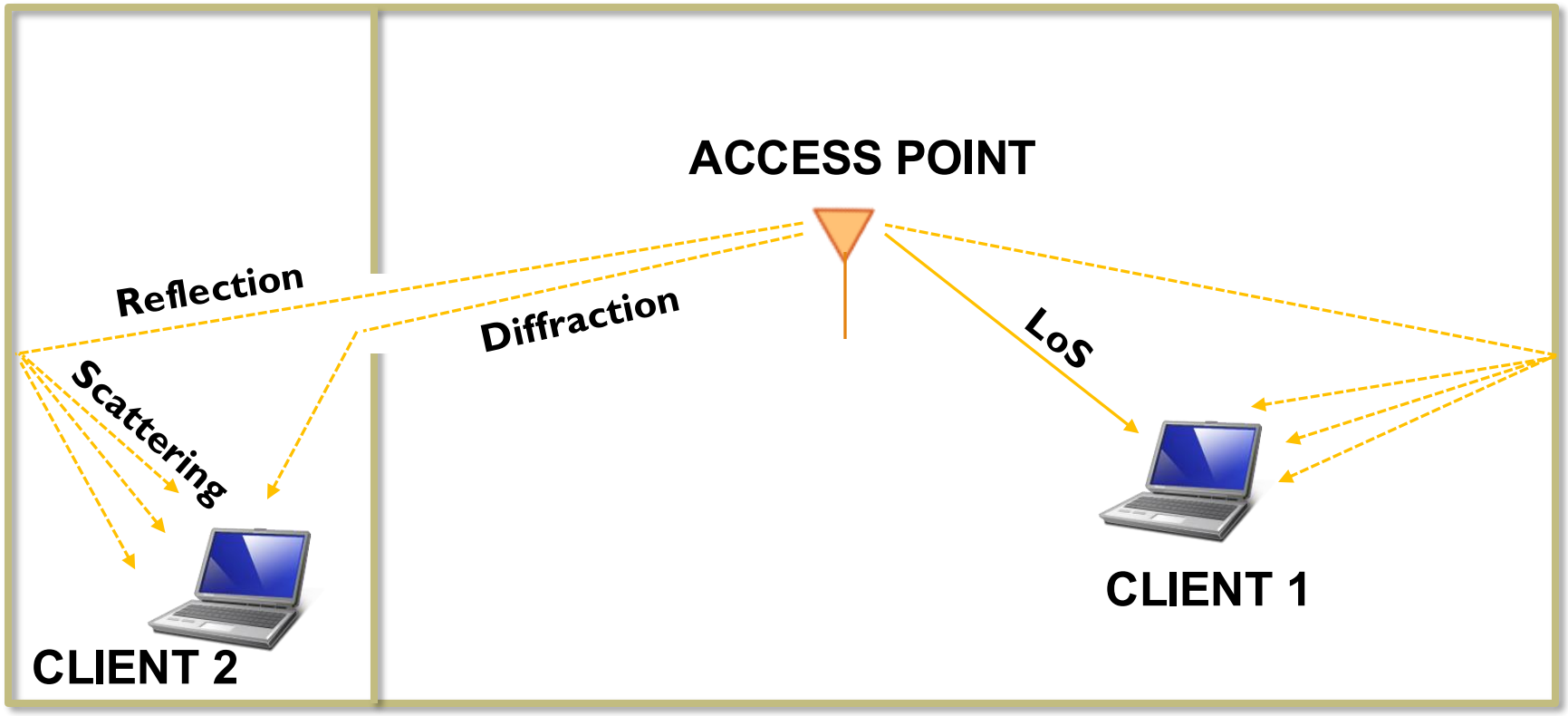
- Soft decision is used for sequence detection, combined with FEC

Analog front end

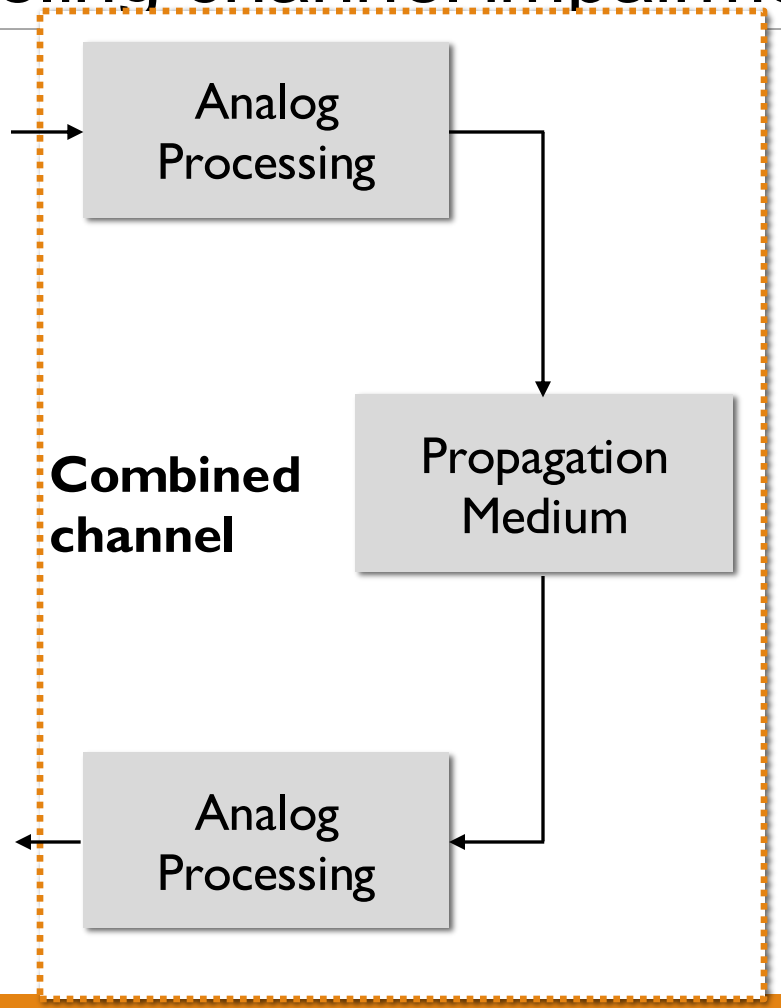


- Includes all processing done in the analog and mixed signal

Propagation effects



Modeling channel impairments



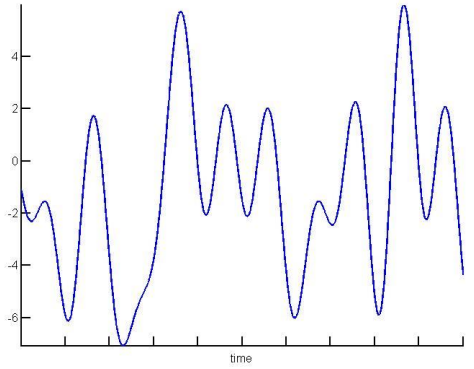
- Combined channel is a model for the combined effects of all the distortions
  - Often called the channel
- The term channel may also refer to a model for just part of the combined channel

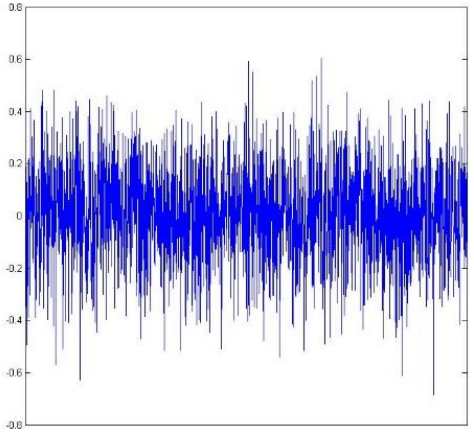
Channel Impairments

Additive noise, normally due to therma electrons

SNR is signal power/noise power

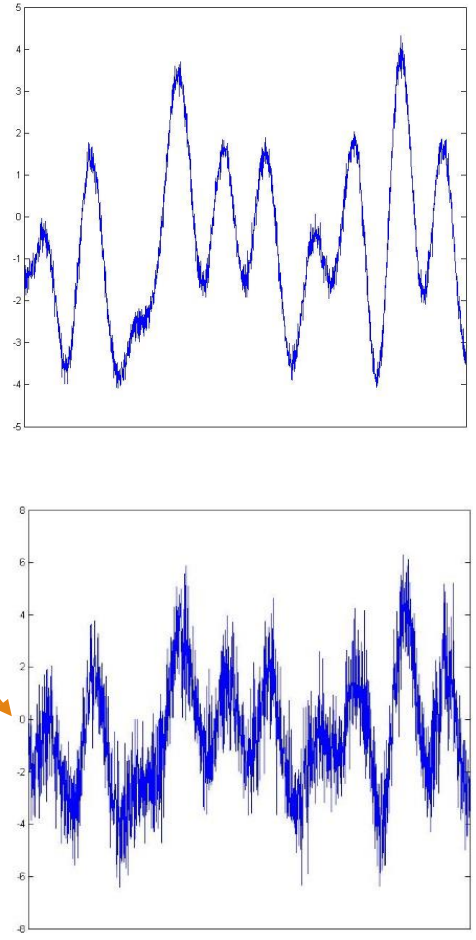
$$y(t) = x(t) + v(t)$$



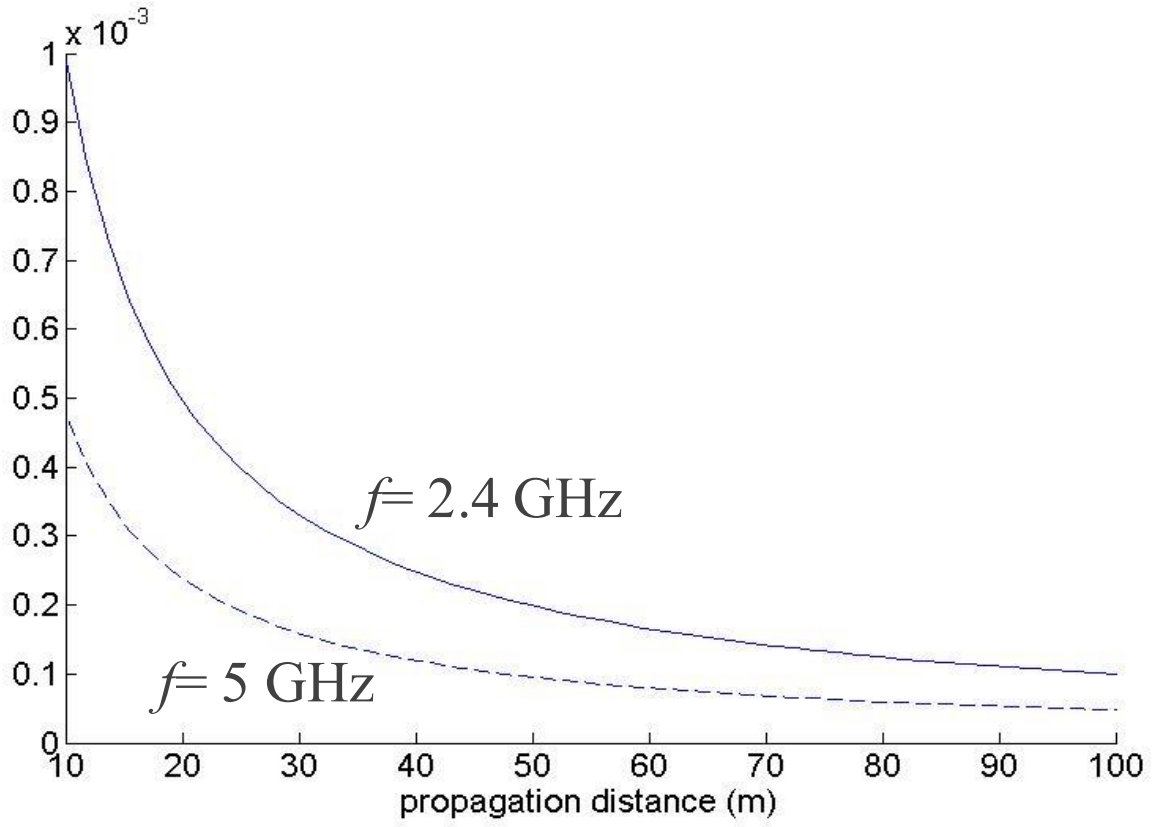


High SNR

Low SNR



Channel Impairments



$\sqrt{G}$

$$y(t) = \sqrt{G}x(t) + v(t)$$

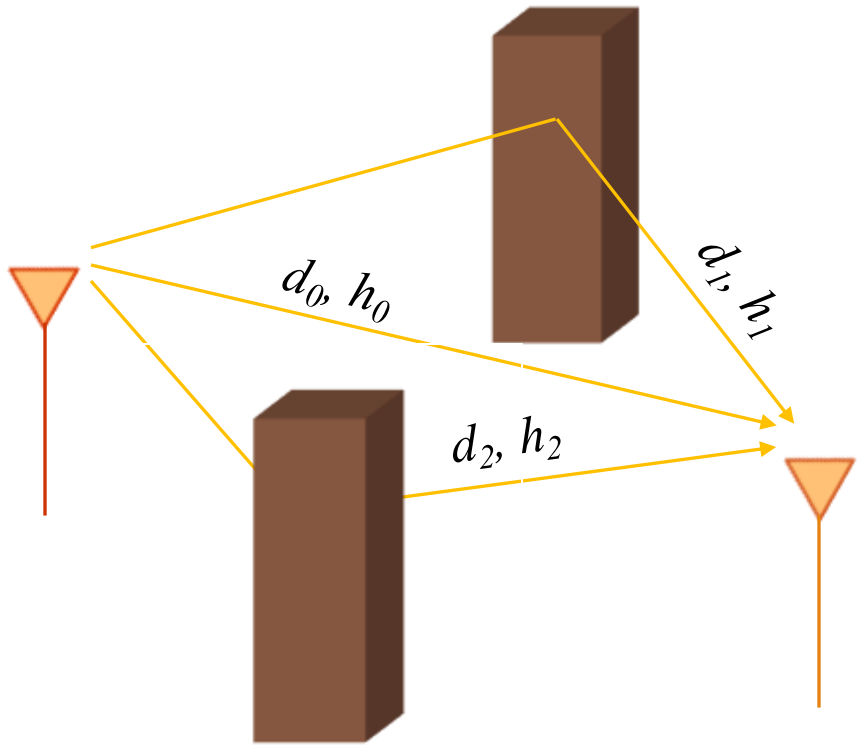
creates an attenuation of the desired signal

noise is added in the AFE, is quite small but not relative to the attenuated signal

Path loss models the degredation of the received signal w/ distance

Channel Impairments

$$y(t) = h_0 x(t - d_0/c) + h_1 x(t - d_1/c) + h_2 x(t - d_2/c) + v(t)$$



- h0 , h1, h2: gains for the existing paths
- d0, d1, d2: distances travelled by the wave from Tx to Rx
- c: speed of light
- v(t): noise

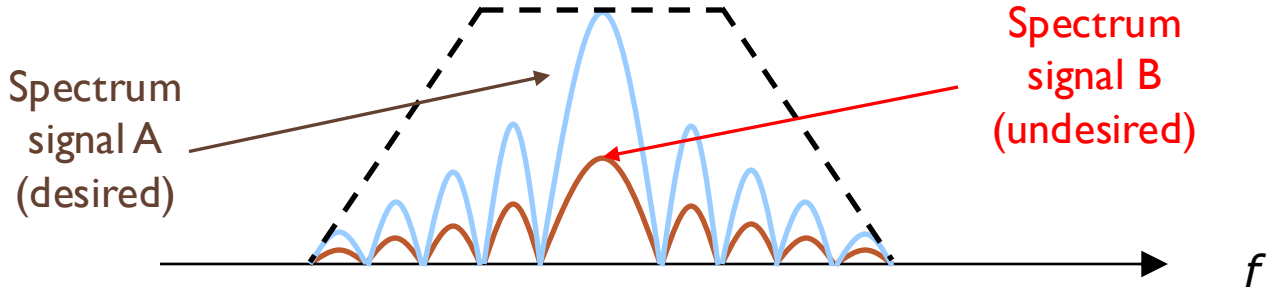
modeled in general as an LTI system

$$y(t) = \int_0^L h(\tau) x(t - \tau) d\tau + v(t)$$

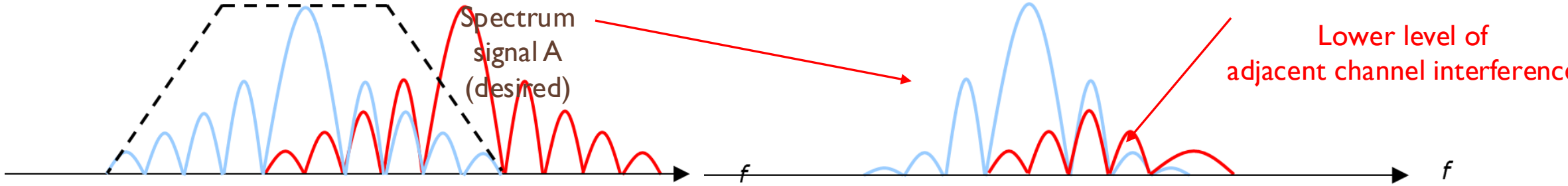
Channel Impairments

Cochannel interference, due to frequency reuse

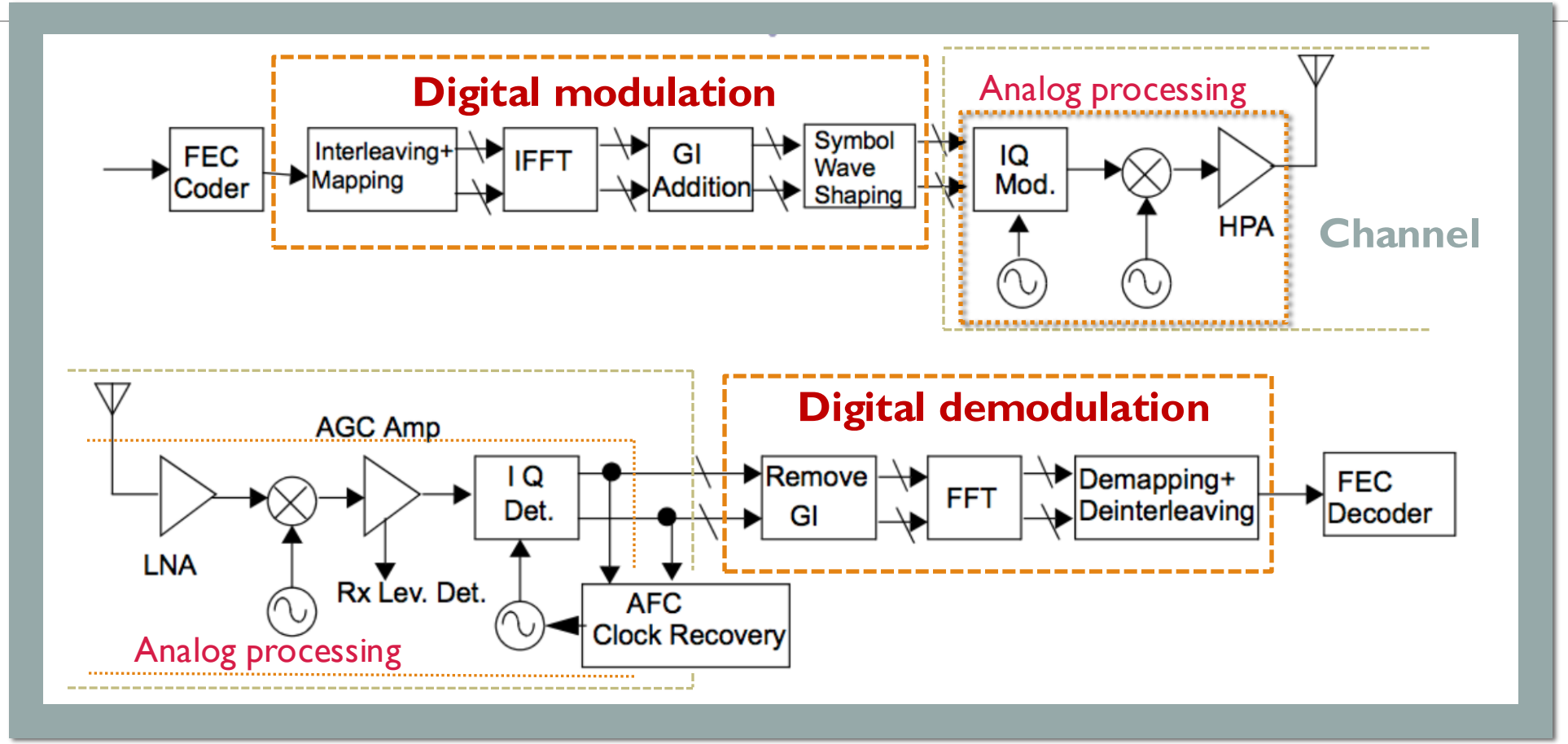
High level of
adjacent channel interference



Adjacent channel interference from spectral leakage



Practical transmitter and receiver block diagram



- OFDM PHY, IEEE 802.11a standard

Summary

Digital communication is widely used in wireless systems

- It has many features that make it more attractive than analog
- Still involves analog processing

There are many critical components in digital communication

- Source encoding / decoding
- Encryption / decryption
- Channel encoding / decoding
- Modulation / demodulation
- Analog-front-end and the propagation channel